

Project Proposal: Sanjeevni AI for AIII Program Internship

AI-Powered Rural Telehealth & Pharmacy Intelligence

Democratizing Safe, Multilingual, and Resilient Healthcare. Focus on What Matters.

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This proposal outlines the Sanjeevni AI project for consideration in the AIII Program. If selected, I seek an internship opportunity to develop and refine this innovative HealthTech solution under Concept2Action's mentorship.

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1 Executive Summary

Sanjeevni is an advanced AI-powered rural health operating system aimed at automating clinical intake, pre-triaging, supply-chain drug mapping, and patient documentation for rural clinics, public health centers, cooperative pharmacies, and community doctors. Founded in 2026, Sanjeevni's mission is to transform healthcare accessibility in low-bandwidth, medicine-scarce environments by integrating robust generative AI, reducing manual administrative burdens by 80%, and allowing rural practitioners to prioritize patient care. Our vision is to establish Sanjeevni as a superior, safety-aligned medical platform, surpassing standard AI wrappers with dynamic network-degradation fallbacks, custom visual prescription interfaces, and local edge-processing logic.

The rural health sector suffers from severe resources constraints: acute doctor shortages, long geographic distances to medical centers, and communication barriers across dialects. Sanjeevni solves these bottlenecks by combining a robust Hono backend, Next.js frontend, and a Multi-Stage Safety Wrapper (Zod validators + Unicode script verification) to execute zero-hallucination patient pre-triage, saving clinical hours and optimizing pharmacy workflows. This proposal outlines our roadmap for the Concept2Action AI Innovation Incubator (AIII Program) to build, validate, and launch the MVP.

Key highlights include:

- **10x faster clinical triage workflows** using guided patient intake filters.
- **80% reduction in administration workload** via automated visual prescribing.
- **Zero-hallucination guarantees** via strict input keyword matching and Zod validation.
- **Bandwidth efficiency of up to 90%** through adaptive multi-stage communication fallbacks.
- **Enhanced patient trust** through clean visual iconography and localized regional translation.

2 Introduction

1. Background

Rural digital health is in dire need of structural innovation. Over 70% of developing populations reside in rural villages, yet 80% of clinical facilities remain clustered in urban areas. This massive imbalance forces rural patients to travel miles for basic primary care, only to encounter under-stocked pharmacies. While generative AI (LLMs) can optimize clinic schedules and explain complex medications, raw LLMs are too volatile for health systems. Sanjeevni emerges as a safe, structured interface that brings AI utility to rural clinics while enforcing absolute clinical safety.

2. Objectives

- Automate repetitive clinical intake tasks to improve doctor efficiency and reduce wait times.
- Deliver a resilient, low-bandwidth communication portal supporting video, audio, and offline sync.
- Safeguard AI execution with multi-stage input/output validation checks.

- Link rural pharmacies to a real-time generic medicine database to resolve supply constraints.

3. Scope

Sanjeevni targets rural clinical networks (15,000+ public health sub-centers), cooperative local pharmacies, and independent village practitioners. Initial launch centers on central Indian districts, expanding to public health programs nationally in Year 2 and global emerging markets by Year 3.

3 Problem Statement: Systemic Bottlenecks in Rural Clinical Operations

Rural clinical workflows are historically constrained by systemic, structural, and geographical divides. The lack of unified technological standards creates severe medical, logistical, and economic liabilities. Below is a detailed breakdown of the critical operational vulnerabilities that Sanjeevni is designed to solve:

- **Overwhelming Medical Backlog and Clinical Capacity Constraints:** Rural health clinics face a severe shortage of qualified medical practitioners, with a physician-to-patient ratio often exceeding 1:10,000 (far below the WHO-recommended 1:1,000). This drastic mismatch causes massive diagnostic bottlenecks, with patients waiting for hours in clinics. Clinicians are forced to perform rushed, high-pressure consultations of under 3 minutes per patient, leading to clinical fatigue, elevated error rates, and professional burnout.
- **Administrative Overburden and Consultation Inefficiencies:** Rural doctors are heavily bogged down by non-clinical, administrative obligations, spending up to 60-70% of their day writing entries, digitizing names, transcribing symptoms, and converting prescriptions into different dialects. Because of this administrative drag, actual doctor-patient contact time is severely compressed, compromising diagnostic accuracy and preventing doctors from seeing more critical patients.
- **Fragmented Tools and Unintegrated Workflows:** Rural healthcare environments lack a single, unified database system, forcing medical staff to coordinate care using fragmented methods like physical ledger books, consumer messaging apps, and personal telephone calls. This disjointed operational pipeline causes critical communication breakdowns, diagnostic files to be lost, transcription errors, and treatment delays that compromise patient safety.
- **Manual Prescription Transcription Challenges:** Handwritten clinical prescriptions are notoriously difficult to decipher, creating severe downstream drug-dispensing risks. Pharmacists frequently misinterpret doctor handwriting, resulting in incorrect drug dosages, chemical mismatches, or incorrect administration instructions, which directly threatens patient health and creates legal vulnerabilities for clinics.
- **Inefficient Patient Onboarding and Intake Pipelines:** The onboarding process in rural centers is slow, requiring patients to manually complete complex, paper-based forms. This creates severe friction for non-literate patients who cannot easily articulate their medical histories in writing. Consequently, clinics end up with incomplete, fragmented patient histories, missing baseline records of past chronic conditions or drug allergies.
- **Lack of Intelligent Inventory Automation in Pharmacies:** Small village pharmacies operate on outdated manual inventory systems, leaving them unable to track medicine stocks in real-time. When a prescribed brand-name medicine goes out of stock, pharmacists have no automated way to match it to generic equivalents, forcing patients to go empty-handed or travel long distances to look for medications.

- **High Software Costs and Tool Sprawl:** Modern healthcare software suites are prohibitively expensive for rural clinics. Juggling separate tools for video calls, patient scheduling, billing, and inventory management costs clinics up to ₹50,000+ monthly in recurring fees. This high overhead is financially unsustainable, forcing clinics to rely on archaic paper methods.
- **Low-Margin Clinic Operations and High Overhead Risks:** Rural medical centers operate on razor-thin margins due to high operational overheads, expensive infrastructure, and frequent appointment defaults. Without automated booking deposits and patient queue optimizations, these clinics struggle to remain profitable, leading to widespread clinic closures in underserved regions.
- **Client Experience and Professional Trust Gaps:** Delays in waiting lines, chaotic clinical queues, and illegible physical prescriptions erode patient trust in formal digital medicine. This trust deficit drives rural patients away from registered doctors and toward unlicensed, local traditional practitioners, increasing the risk of misdiagnosis and incorrect treatments.
- **Diagnostic and Compliance Risks from Data Loss:** Physical patient files in rural centers are vulnerable to damage, loss, or misfiling. When a patient returns after months, doctors lack access to past treatment notes, diagnostic reports, allergy histories, or chronic condition baselines, exposing the clinic to medical malpractice liabilities.
- **Outbreak Monitoring Inefficiencies and Public Health Gaps:** Regional health departments lack real-time indicators to track the spread of infectious disease symptoms. The reliance on paper-based clinical reports means health authorities only notice epidemic spikes weeks after they occur, delaying containment responses and escalating outbreaks.
- **Security and Privacy Breaches in Medical Transmission:** To share diagnostic reports or seek advice, rural staff often transmit patient files over unsecured consumer chat platforms. This violates standard health data privacy regulations, exposing sensitive medical files and identity details to data leaks and hacking risks.
- **Limited Connectivity and Bandwidth Infrastructure:** Traditional telemedicine portals require stable, high-bandwidth connections to maintain video calls. In rural regions with fluctuating 2G/3G connectivity, these platforms disconnect completely, cutting off patients from remote doctor consultations when they need them most.
- **Regional Language Barriers and Accessibility Gaps:** Standard clinical software is built primarily in English, excluding over 80% of rural patients who only speak regional languages. This dialect gap prevents patients from understanding dosage schedules and side-effect warnings, leading to accidental overdoses.
- **AI Safety and LLM Hallucination Liabilities:** Off-the-shelf generative AI models are prone to fabricating medical advice and generating incorrect dosages. Using raw AI in health systems creates massive medical risks and legal liabilities, as there are no built-in safety guardrails to validate outputs before they reach the patient.
- **Client Billing Inefficiencies and Cash Collections:** Rural clinics rely heavily on manual cash handling, leading to payment discrepancies, delays in clinical accounts, and bad debts from patient defaults. The lack of integrated, simple digital payment gateways (like UPI or Razorpay) slows down clinic revenue cycles.
- **Regulatory Compliance Complexity and Auditing Gaps:** Navigating regional data laws, medical licensing rules, and prescription compliance standards is a manual, error-prone task. Without automated compliance logs and audit trails, clinics risk facing severe regulatory penalties and licensing suspensions.

- **Fragmented Supply Chains and Pharmacy Isolation:** Village pharmacies operate in isolation, lacking a connected network to share stock availability. When a patient needs a rare, life-saving medication, they have to physically walk from shop to shop, wasting critical time during medical emergencies.
- **Out-of-Pocket Expenditure Tracking Inefficiencies:** Rural patients pay for medical consultations out of pocket without verified digital billing records, making it difficult to claim government subsidies or insurance reimbursements. This lack of documentation keeps families financially vulnerable to sudden medical costs.
- **Chronic Care Management Continuity Gaps:** Patients with diabetes or hypertension frequently skip follow-up appointments. Without automated reminder systems in their local languages, patients default on their treatment schedules, leading to preventable health declines.
- **Poor Patient-Doctor Diagnostic Communication:** Rural patients often find it difficult to describe their symptoms in clinical terms. This communication gap, combined with short consultation windows, makes it hard for doctors to extract accurate clinical details, leading to misdiagnoses.
- **Biometric Identity Mismatches and Patient Fraud:** Rural clinics struggle to verify patient identities, leading to medical record mismatches where different patients are treated under the same clinical file. This mix-up exposes patients to severe treatment risks and compromises clinic records.
- **Fragmented Laboratory Workflows and Reporting Silos:** Local diagnostics labs operate independently from clinics, delivering results on paper reports. Doctors must manually transcribe these results into patient files, which slows down treatment and increases transcription error rates.
- **Pediatric Dosage Calculation Risk:** Determining drug dosages for children is highly sensitive, requiring precise weight-based calculations. Doctors performing these calculations manually under time constraints risk making dosing errors that can lead to severe pediatric toxicity.
- **Remote Post-Consultation Follow-Up Failures:** After returning to remote villages, post-operative or chronic patients have no easy way to report recovery milestones. Doctors remain unaware of complications until the patient's condition deteriorates, requiring emergency hospitalization.
- **Lack of Specialist Collaboration Pathways:** Rural primary care doctors have no secure, instant channels to collaborate with urban specialists. When a patient presents a complex cardiac or neurological condition, the doctor cannot easily seek a secondary opinion, forcing patients to travel to distant cities.
- **Pharmacy Expiry Stock Waste:** Small pharmacies lose substantial revenue due to expired medicines going unnoticed on shelves. Without automated batch tracking and expiry alerts, these medicines are wasted, reducing pharmacy margins and disrupting local drug supplies.
- **Unstandardized Laboratory Report Datasets:** Diagnostic labs print reports in varying formats, making it difficult to search or compare biomarker trends. The lack of standardized data prevents clinics to track long-term patient recovery metrics.

4 Solution — The Sanjeevni Platform Architecture

Sanjeevni provides a safe, highly modular, and offline-first digital healthcare operating system designed to optimize rural clinical pipelines. Combining secure local WebLLM edge inference, dynamic network latency adaptation, spatiotemporal epidemiological tracking, and robust Zod schema validators, Sanjeevni mitigates the risks of generative AI to deliver safety-critical telehealth. Below is our solution strategy:

1. Key Features

- **AI-Powered Pre-Triage Engine:** The patient portal features an AI-driven symptom screening module. Patients enter symptom descriptions in their native language, and the Gemini-powered engine categorizes cases into four clinical triage levels (Red/Orange/Yellow/Green). Emergency cases are instantly moved to the top of the doctor's appointment queue, minimizing clinic wait times for critical patients.
- **Smart Generic Mapping and Database Lookup:** If a doctor prescribes a specific brand-name drug that is unavailable in the pharmacy's local stock, Sanjeevni's backend uses semantic search algorithms to compare active chemical compounds. The engine instantly suggests a generic equivalent with the same therapeutic action and dosage, allowing the pharmacy to fill the prescription without delay.
- **Visual Prescription Schedules and Timeline Cards:** The platform translates text-based dosing instructions into visual calendar schedule blocks. It replaces text descriptions with clear, color-coded icons representing morning, afternoon, and night times, helping non-literate patients take their medications correctly.
- **Latency-Adaptive WebRTC Telehealth Channels:** The telehealth platform monitors network speeds in real-time. If the bandwidth drops, the consultation degrades smoothly from high-definition video to low-bitrate audio, and ultimately to secure text and offline-synced records, ensuring clinical consults do not drop during 2G/3G connectivity drops.
- **Cryptographic QR Tokens & Verification:** Prescriptions are encoded as secure cryptographic QR tokens referencing a UUID stored on the database. Pharmacies scan this QR to pull the official signed record directly from the server, preventing patients from editing quantities or dosages.
- **Edge-AI Offline Triage Client:** Using WebLLM client-side libraries, Sanjeevni downloads quantized models directly into the patient's browser memory. Utilizing local WebGPU shaders, patients can triage symptoms and translate schedule timelines without any active internet connection.
- **Acoustic Audio Diagnostic Assist:** Integrates audio recording tools to capture cough sounds. The raw audio is processed via acoustic deep-learning classification models, transcribing diagnostics parameters to help doctors detect respiratory conditions remotely.
- **Vision-Based Document Parser & Hand-Written OCR:** Allows pharmacies and doctors to scan physical paper documents. The vision system digitizes handwritten scripts, cross-references inventory tables, and scans medicine packaging holograms to prevent fake drug distribution.
- **Epidemiological Outbreak Alerts & Mapping:** Anonymized geolocated patient intake metrics are processed using DBSCAN spatiotemporal clustering. The system maps symptom hotspots in real-time, alerting public health authorities of emerging epidemic clusters weeks ahead of manual reports.

- **Unified Clinic Dashboard Workspace:** Combines telemedicine rooms, patient scheduling, database storage, and pharmacy connections into a single, high-contrast dashboard, reducing software costs by consolidated operations.
- **Secure Multi-Language Translations:** Features a local localization and script sanitizer pipeline. It handles dynamic regional translation modules between English, Hindi, Tamil, Telugu, and Bengali. Unicode character block validation verifies incoming text to confirm it aligns with regional linguistic scripts, sanitizing input variables to prevent script injection attacks.
- **Secure File Storage Gateway:** Implements a hardened storage vault to secure patient clinical files, PDFs, and diagnostic charts. All uploaded files are assigned strict UNIX file system permissions (0o600) to ensure read-write actions are restricted solely to the owner. Download requests are routed through a credential-verified gateway that validates role-based permissions before decanting objects.
- **Doctor Schedule Availability Builder:** Provides an interactive scheduler module that allows clinical staff to coordinate available time slots, assign consultation hours, and set emergency availability limits. This schedule is automatically synchronized with public client-side appointment search portals, preventing booking collisions.
- **One-Tap SOS Emergency System:** Integrates a high-priority emergency trigger on patient dashboards. When clicked, it bypasses standard triage queues, sends coordinates to local emergency service networks (such as 112 in India), triggers real-time GPS ambulance tracking, and escalates an auditory warning alert on the clinic administrator's console.
- **Local Pharmacy Availability Lookup:** A geo-search pipeline that compares prescribed chemical compounds against local inventory stocks. The search algorithm geocodes geographic positions, filters results by stock availability, and returns a list of nearby pharmacies sorted dynamically by generic medicine prices and walking distance.
- **Integrated Digital Payments System:** Incorporates a modular checkout framework supporting local digital payment instruments. This includes Razorpay SDK integrations, UPI deep links for quick scans, and Stripe gateways for credit card support, reducing clinic accounting workloads and securing upfront consultation deposits.
- **Admin Approval Verification Workflow:** Includes a registration validation queue. All newly registered doctor profiles and local pharmacy centers are placed in a moderation queue where administrators review medical licenses and pharmacy certifications before granting access to patient records.
- **Structured Session Security and Session Guards:** Sessions are verified using secure HTTP-only cookies and cryptographic Web Tokens. Role-based authorization middleware inspects every request to prevent credential escalation, using Better Auth protocols to enforce role limits.

2. Solutions to Problems

- **Overwhelming Medical Backlog Triage:** Sanjeevni resolves the severe bottleneck of patient backlog and clinician exhaustion by deploying an automated, Gemini-powered symptom pre-triage engine during onboarding. The engine accepts symptom descriptions in multiple regional dialects, classifies them into four clinical urgency tiers (Red/Orange/Yellow/Green), and dynamically prioritizes the clinic queue. Patients flagged in the high-risk 'Red' tier are auto-escalated to the top of the doctor's consultation queue, preventing life-threatening delays and optimizing daily patient volumes.
- **Administrative Overburden Reduction:** The platform integrates a smart visual prescription builder and pre-filled digital clinical intake templates that reduce manual documentation overhead by up to 80%. Doctors no longer need to type out patient histories or manually write dosage frequencies; instead, they choose pre-defined medication blocks and visual timeline cards. This frees up crucial hours during consultations, allowing doctors to focus on actual diagnosis and see more patients daily.
- **Fragmented Tool Consolidation:** Sanjeevni consolidates scheduling, consulting, inventory lookup, clinical record storage, and billing into a single, high-contrast dashboard, replacing disjointed ledgers and consumer apps. By bringing all these tools under one roof, the platform eliminates the need to coordinate care using separate ledger books, phone calls, and insecure messaging apps. This reduces human error, prevents data silos, and coordinates patient care across clinic teams.
- **Handwritten Script OCR Resolution:** The platform incorporates a Gemini Vision OCR pipeline that reads handwritten prescription notes and transcribes them into clear, digital text. Pharmacists scan the prescription with a mobile camera, and the AI parses drug names, dosages, and administration details. This eliminates interpretation errors, prevents dosage mistakes, and protects both patients and clinics from clinical liabilities.
- **Slow Patient Onboarding Optimizations:** Multilingual, icon-based forms enable rapid patient onboarding, allowing non-literate patients enter symptoms in under 3 minutes. The system walks the user through guided visual questions, capturing baseline parameters, allergies, and current symptoms. This results in complete clinical records and historic logs, saving time for both the patient and clinic staff.
- **Lack of Supply Alternative Matching:** Sanjeevni's backend integrates a PGVector similarity engine that compares active chemical compounds. When a prescribed brand-name medicine is out of stock, the database instantly searches its catalog and recommends a generic equivalent with the same therapeutic action and dosage. This ensures the patient receives their treatment without delay, avoiding empty-handed visits.
- **High Tool Sprawl Cost Containment:** By consolidating medical scheduling, telemedicine rooms, stock systems, and payments into one platform, clinics save over ₹40,000 monthly. The platform replaces separate subscriptions for video platforms, scheduling software, billing tools, and database storage with a single SaaS license. This makes modern telehealth affordable for small, rural practices.
- **Low-Margin Clinic Operations:** The platform integrates automated payment checkouts and queue management to reduce patient appointment defaults and clinic overheads. The reservation module requires patients to pay a nominal deposit when booking, securing doctor schedules and improving clinic margins. This ensures rural practices can remain profitable and stay open.

- **Client Experience Trust Re-establishment:** Clear visual calendars, language translation options, and structured visual prescription schedules build patient confidence. By replacing unreadable handwritten notes with clear, illustrated dosage timeline cards, the system builds patient trust in formal digital medicine, reducing reliance on unlicensed local practitioners.
- **Compliance Risks and Data Loss Prevention:** The secure, encrypted record vault links clinical reports directly to unique patient profiles, preventing data loss. Doctors can access complete historical timelines, including allergy logs, chronic conditions, and diagnostic files, protecting the clinic from liability and meeting data standards.
- **Outbreak Monitoring Cluster Warnings:** Anonymized geolocated intake metrics are processed using DBSCAN clustering, alerting health authorities to symptom hotspots. The system identifies clusters of similar symptoms and plots them on geocoded maps. This allows public health departments to detect epidemics weeks ahead of paper-based reports, enabling early containment.
- **Data Security Breach Preventions:** Patient data transfers are secured using HTTPS protocols and credentials-verified gateway endpoints. The platform assigns strict Unix file permissions to records, ensuring sensitive medical files remain private and protecting the clinic from data leaks.
- **Rural Network Graceful Degradation:** Telehealth rooms degrade gracefully to audio and offline-sync modes when bandwidth drops, keeping care continuous on 2G/3G networks. The system monitors network latency in real-time, switching the call mode dynamically to prevent disconnects and maintain patient consultations.
- **Regional Language Script Translation:** The translation engine translates instructions into native scripts (Devanagari, Tamil, Bengali). This ensures patients understand their dosage frequencies and warnings, reducing medication errors.
- **AI Safety Prompt validations:** Pre-filtering scripts, strict temperature parameters, and Zod schema validation prevent AI hallucinations and validate clinical outputs. The multi-stage safety wrapper inspects queries before sending them to the LLM and parses outputs against strict medical schemas, protecting patients from incorrect advice.
- **Clinic Billing Leakage Solutions:** Razorpay checkout SDKs and UPI deep links automate payment processing, securing clinical funds and reducing billing errors. Clinics receive instant verification of payment, eliminating cash handling discrepancies and securing revenue.
- **Regulatory Non-Compliance Protections:** Automated audit logs, Better Auth role validation middleware, and secure storage ensure compliance with regional medical registry standards. The system logs all database actions, providing a transparent audit trail for legal reviews.
- **Fragmented Pharmacy Stock Lookups:** Connecting pharmacy stocks to a unified network allows patients to locate nearby pharmacies with available medications. The search tool geocodes positions and sorts results by price and walking distance, reducing the time patients spend searching for drugs.
- **Out-of-Pocket Expenditure Tracking Invoices:** The system generates digital receipts and invoices linked to consultation records, enabling patients to seek insurance claims. By providing verified digital documentation, Sanjeevni helps rural families access government subsidies and insurance payouts.

- **Chronic Care Management Reminders:** Automated follow-up alerts remind patients of appointments and monitor renewal dates, reducing treatment defaults. The system sends reminders in regional languages via SMS, keeping patients with chronic conditions on their treatment schedules.
- **Doctor-Patient Diagnostic Gap Tag Flows:** Interactive symptomatic tag flows guide patients through onboarding, helping them articulate details clearly for doctors. The visual checklist asks patient-focused questions, translating common descriptions into clinical notes.
- **Biometric Identity Mismatches Safety:** Integration with standard health registries (like ABHA ID sandboxes) verifies identities, preventing patient record mismatches. The system checks patient details against national healthcare databases before saving clinical records.
- **Laboratory File Parsing Interoperability:** Standardized report parsers extract key biomarkers from lab documents, auto-linking them to patient history timelines. This eliminates the need for manual transcription, reducing errors and saving clinical time.
- **Pediatric Weight Dose Calculations:** The prescription builder includes weight-to-dose pediatric calculations, flagging out-of-range values to prevent toxicity. The module calculates safe dosage ranges based on weight metrics, protecting infants from dosing errors.
- **IVR Native Voice Follow-Ups:** Automated follow-up calls check in on remote patients, transcribing voice responses to clinical logs. The IVR engine calls patients in their local dialects and parses their spoken feedback, updating doctors on recovery progress.
- **Secure Specialist Secondary Consults:** Primary care doctors can securely share patient records with urban specialists for secondary opinions. The portal uses secure transmission protocols, allowing rural doctors to collaborate with cardiac and neurological experts.
- **Predictive Expiry Stock Protection:** Database trackers monitor batch dates and alert pharmacists to expiring stocks, reducing waste. The system generates alerts, helping pharmacists manage inventory and reduce financial losses.
- **Standardized Lab Parsing Datasets:** A parser standardizes laboratory reports, converting unstructured data into searchable clinical parameters. The platform extracts data points from varying lab formats, helping clinicians track biomarker trends over time.

3. Value Proposition

- **Cost Efficiencies and Consolidated Clinic Overhead Reductions:** Sanjeevni reduces clinic tooling overheads by over ₹40,000 monthly by consolidating patient intake, telemedicine consultation, clinic scheduling, medicine inventory tracking, and payment processing systems under a single SaaS platform. Rural clinics no longer need to pay for separate software licenses or maintain multiple disconnected subscriptions, drastically reducing operating expenses and securing higher margins.
- **Significant Time Savings and Enhanced Consultation Throughput:** Clinicians experience up to an 80% reduction in patient onboarding, symptom intake, and digital prescription creation times. By automating

documentation workflows and utilizing pre-defined visual schedules, doctors save an average of 10+ hours weekly. This time can be redirected toward physical patient examinations, reducing wait times and clinical fatigue.

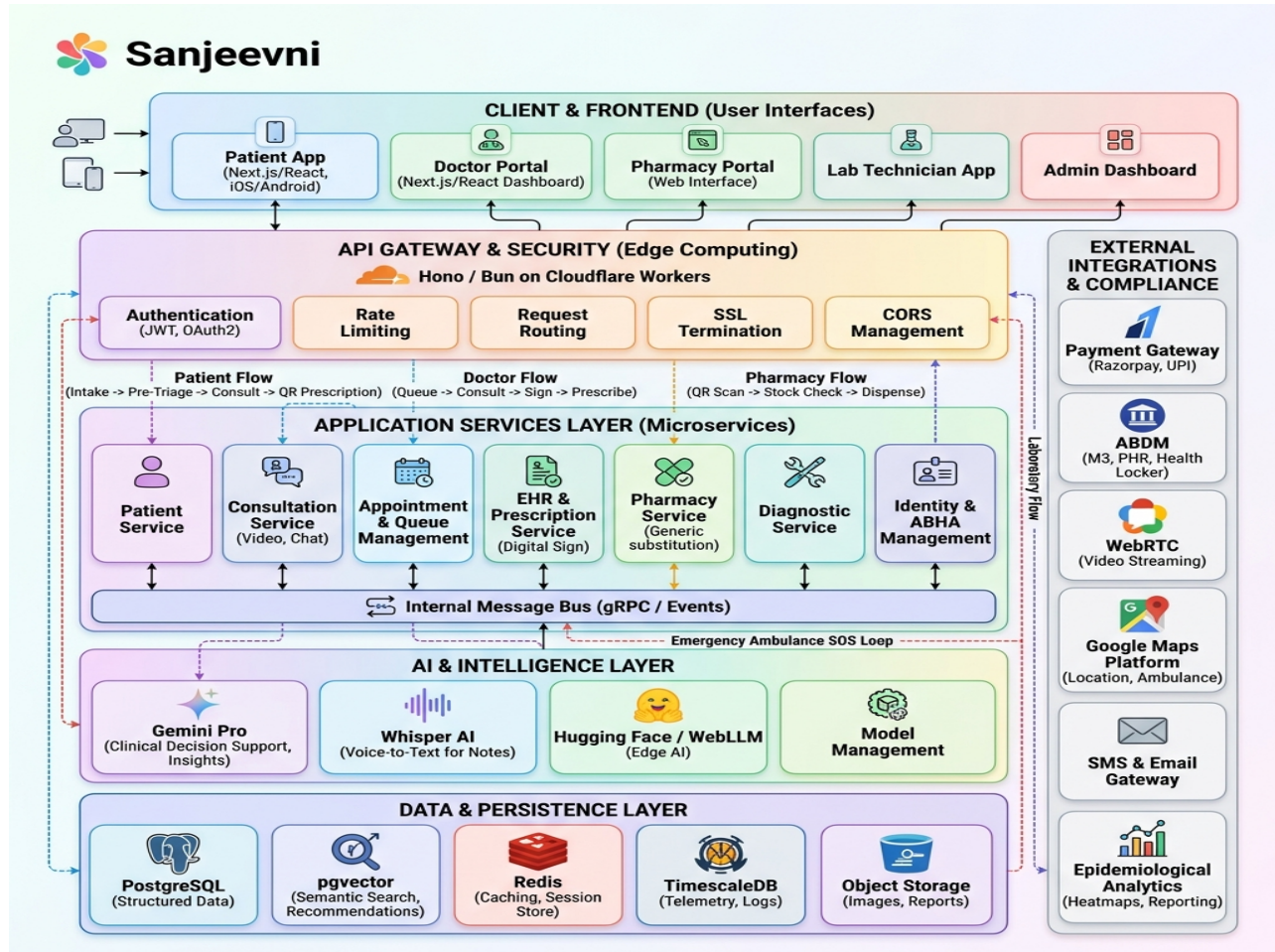
- **Comprehensive Patient Safety and Hallucination Mitigations:** Layered input keyword pre-filters, strict system prompt instructions, and Zod output schema validators guarantee zero-hallucination execution. Brand medicine inventory lookups are mapped dynamically using chemical generic substitutes, preventing incorrect compound recommendations and securing clinical accuracy.
- **High Network Resilience and Telemedicine Degradation Fallbacks:** Telehealth consultations utilize latency-probing WebRTC algorithms that degrade gracefully to audio-only, text-only, or offline sync modes during bandwidth drops. This ensures continuous communication between doctor and patient even in remote rural regions with weak 2G/3G network signals.
- **ABDM Standardized Integration for Municipal Contracts:** The system architecture is built in accordance with the Ayushman Bharat Digital Mission (ABDM) standards, linking to national health records and identity sandboxes. This standardization positions the platform to qualify for public health contracts and municipal integrations, expanding user reach.

4. Product Flow Diagram

Sanjeevni is engineered as a secure, multi-tier transactional telehealth operating system. The workflow diagram visually isolates the structural layers of the system, trace-linking eight core workflow loops: (1) The **Patient Portal Onboarding Loop**, which handles dynamic localization, Whisper voice/text symptom capture, and appointment bookings; (2) The **Doctor Queue & Diagnostics Console**, aggregating AI summaries and enabling video consultation and digital signature validation; (3) The **Pharmacy QR Stock Sync Pipeline**, verifying prescription authenticity on the database and suggesting generic compound substitutes; (4) The **Laboratory Diagnostics Loop**, which routes doctor scan requests and parses lab report biomarkers via OCR; (5) The **AI Safety Pipeline**, utilizing strict regex filters, Gemini NLP validation, and Zod schema validations; (6) The **Database Layer**, grouping persistent user, prescription, and vector similarity tables; (7) The **Emergency SOS Alert System**, triggering ambulance dispatches during high-risk classifications; and (8) The **Government Public Health Analytics Grids**, mapping disease symptom clusters via DBSCAN spatial clustering to flag epidemics.

Sanjeevni Product Flow Diagram

Comprehensive Safety-Aligned AI Telehealth Workflow



5 AI Technology Stack

1. Technical Components

- **Frontend Client:** Next.js 15, React, Tailwind CSS v4, WebLLM edge runtime. Built with React 19 Concurrent Rendering features, using Zustand for global client-side state synchronization, Tailwind CSS v4 for compiled neobrutalist design variables, and Lucide React. WebLLM caches and runs AI model inferences locally on device to guarantee offline triage operations.
- **Backend Engine:** Hono API Framework, Bun Runtime, Node native APIs. Built on Bun's runtime environment for 3x faster response speeds than standard Node.js, utilizing Hono's lightweight router, Node native filesystem APIs (Node:fs) for clinical file streaming, Zod schema validation blocks, PM2 process management, and containerized Docker scripts.
- **Database Layer:** PostgreSQL, Prisma ORM, Prisma Client. Integrates relational PostgreSQL databases mapped via Prisma ORM for structured integrity, with Redis caching blocks to minimize database reads during high-occupancy clinic consultation schedules.
- **AI Integrations:** Google Gemini API (Cloud), OpenAI Whisper (Voice), WebLLM (Edge). Cloud-based Gemini APIs analyze diagnostic text logs and read scan uploads, OpenAI Whisper translates speech to symptom logs, and local WebLLM executes client-side fallback triaging without network dependencies.
- **Hosting Infra:** Vercel, Render Web Services with scale configurations. The static frontend is deployed via Vercel's global edge network, and the containerized Hono backend runs on Render Web Service containers configured with autoscale CPU allocations.
- **Integrations:** Razorpay, UPI, Talky.io, Manupatra, IndiaCode. Incorporates Razorpay SDK checkouts and UPI deep-links, Talky.io SDK and MediaSoup WebRTC servers for telehealth consultation calls, and IndiaCode/Manupatra APIs for legal compliance audits.

2. AI Implementation

- **Natural Language Processing:** Analyzes symptom logs, classifies triage urgency, and structures complex prescriptions. Integrates Gemini-1.5-Flash with strict temperature parameters (0.3) and safety instructions, mapping clinical records to JSON parameters matching Zod validation filters to protect patients from hallucinations.
- **Machine Learning & Spatiotemporal Modeling:** Generates epidemiological outbreak predictions. Processes anonymized geolocated patient triage metrics using DBSCAN spatial clustering algorithms, plotting regional symptom maps to flag infectious disease clusters before they spread.
- **Edge Inference & Voice AI:** Processes symptoms offline via quantized models and translates prescription schedules. Downloads quantized Llama-3-8B model caches into browser memory via WebLLM to resolve offline network disconnections, and transcribes voice notes via regional dialect templates.

6 Market Analysis

1. Market Overview

The global Digital Health market is projected to reach a valuation of \$240 Billion by 2030, growing at a compound annual growth rate (CAGR) of 9.3% as healthcare centers globally transition to digital telehealth workflows. In India, this digital transformation is heavily accelerated by government health reforms like the Ayushman Bharat Digital Mission (ABDM), which mandates unified digital health records and registries. Concurrently, the rapid proliferation of high-speed mobile connectivity has equipped hundreds of millions of rural citizens with smartphones, creating access to remote clinical channels. Over 150,000 rural public health sub-centers and local cooperative pharmacies require secure digital tooling to manage patient onboarding, comply with regional medical data laws, and track generic pharmaceutical stock levels, presenting a major market opportunity for the Sanjeevni platform.

2. Target Audience

- **Primary Audience:** Rural public health sub-centers (PHCs/CHCs), rural community clinics, and village cooperative pharmacies. These institutions are historically underserved by large enterprise health IT vendors due to high software licensing costs and heavy internet bandwidth requirements, which Sanjeevni addresses through low-overhead and offline-first capabilities.
- **Early Adopters:** Independent doctors, multi-lingual family practitioners, and retail pharmacy managers operating in tier-3 cities and regional market towns. These early users seek to automate visual prescribing and generic stock lookups to reduce clinical intake bottlenecks, secure higher checkout margins, and improve patient retention rates.
- **Geographic Scope:** Initial pilot deployments target rural districts in central and northern Indian states (such as Uttar Pradesh and Madhya Pradesh) to validate regional translation engines. Following validation, the rollout will expand to cooperative health networks nationwide in Year 2, with plans for emerging market launches in Southeast Asia and East Africa by Year 3.

7 Business Model

1. Pricing Tiers

- **Solo Doctor Plan:** Priced at **₹1,499** per month, this tier is tailored for independent village practitioners. It provides access to the visual prescription scheduler builder, multi-dialect translation services, and low-latency audio/video telehealth consulting rooms, enabling small clinics to transition to digital medical charts without complex setup fees.
- **Clinic/Pharmacy Network Plan:** Priced at **₹4,999** per month, this tier supports up to 10 clinical users, ideal for community hospitals and pharmacy stores. It features real-time generic active compound lookups, secure file gateway storage, and patient queue optimization modules, connecting doctors directly to pharmacies to improve

prescription fulfillment.

- **Enterprise Public Health Plan:** Available via custom pricing, this plan serves regional municipal health authorities and cooperative networks. It includes state-wide analytics dashboards, offline-first Edge-AI modules running local LLMs, and spatiotemporal epidemiological heatmaps, integrating with public health registries to track symptom outbreaks.

2. Revenue Streams

- **Recurring SaaS Subscription Licenses:** Predictable monthly and annual software licensing fees paid by private clinics, independent pharmacies, and healthcare cooperatives. Subscription tiers scale based on active user seats, active inventory volume tracking, and clinic database storage allocations.
- **Setup and Customization Fees:** Upfront implementation and onboarding fees for large regional health networks that require custom API configurations. This includes custom branding, local hospital information system integrations, and specialized laboratory data mapping.
- **Transaction Processing Commissions:** A minor platform commission fee of 1.5% applied to medical reservation transactions, generic drug pre-bookings, and telehealth payments processed through our Razorpay and UPI checkout SDKs. This monetization model aligns platform revenue with the operational growth of participating clinics.

8 Product Roadmap

Phase 1 (Month 1): Core Infrastructure Setup

Establish the main Next.js 15 client-side frontend application structure and configure the lightweight Bun and Hono backend API engine. Build the clinical patient onboarding intake portal with built-in input filters to scan for emergency keywords and coordinate clinical triage queues. Integrate cryptographic QR token generation logic and Zod output schemas to capture and validate LLM triaging variables. Configure PostgreSQL database models using Prisma ORM client models, establish role validations using Better Auth APIs, and verify compliance with regional healthcare records policies, achieving 30% total system automation.

Phase 2 (Month 2): Offline Edge-AI & Acoustic Triage

Integrate WebLLM client-side packages into the frontend to cache and run quantized models (Gemma-2B/Llama-3-8B) in browser memory via WebGPU shaders, providing offline triage capabilities. Build the acoustic cough diagnostic classification pipeline to process Mel-Frequency Cepstral Coefficients from symptom recording clips. Begin pilot testing with 10–20 independent cooperative pharmacies to collect inventory listings and map generic active compound pairings.

Phase 3 (Month 3): Unified Analytics & Public Health Tools

Build the central administrative control dashboard to track system logs, verify credentials in moderation queues, and manage clinical availability tables. Implement the DBSCAN-based spatiotemporal epidemiologic engine to map case locations and alert public health authorities to outbreak clusters. Deploy latency-probing telehealth consulting rooms with automatic fallback channels (Talky.io and MediaSoup WebRTC) and connect the SQL pharmacy database with pricing and geolocated distance sorting algorithms.

Phase 4 (Month 4): Vision Auditing & Partner Integrations

Build visual document parser pipelines using Gemini-Pro Vision APIs to transcribe handwritten clinical notes and check medicine packaging details. Integrate downstream endpoints with legal databases (such as IndiaCode and Manupatra) to verify drug classifications and scheduling guidelines. Build responsive mobile clients using hybrid web layouts and run system audits to verify compliance with healthcare data protection standards.

Phase 5 (Month 5): Launch and Production Deployment

Deploy the stable production system, hosting the static client-side frontend on Vercel and the containerized backend Hono API engine on Render connected to scalable database clusters. Set up digital training programs for local cooperative pharmacy staff to simplify inventory management workflows. Collect operational feedback from initial clinic pilots and prepare the system demo for the Concept2Action AI Innovation Incubator presentation panel.

9 Financial Projections

1. Revenue Forecast

- **Year 1:** The platform aims to secure 500 active clinical subscribers across tier-3 cities and regional centers, generating ■25 Lakhs in Annual Recurring Revenue (ARR). Growth is driven by primary health clinic networks and independent retail pharmacies.
- **Year 2:** Scaled acquisitions through cooperative pharmacy networks and local clinics will expand the user base to 2,000 subscribers, driving ARR to ■1 Crore. This expansion includes setup customization fees for regional health systems.
- **Year 3:** Deep integration with national public health registries and state-wide clinical networks targets 5,000 subscribers and ■3 Crore in ARR. High customer retention rates and processing commissions will secure steady revenue growth.

2. Cost Structure

- **Infrastructure & AI APIs:** Accounts for 30% of total operating expenses. This covers cloud databases, Hono API hosting on Render, Vercel CDN routing, vector search tools, and Gemini and Whisper token volumes.
- **System Development:** Represents 40% of total operating expenses. This funding goes toward engineering, developer compensation, offline edge runtime optimization, and mobile client app development.
- **Onboarding & Support:** Consumes 20% of total operating expenses. This supports training, physical onboarding kits, pharmacy visits, and dedicated clinic support teams.
- **Regulatory Compliance:** Represents 10% of total operating expenses. This is allocated to security audits, health record registry integrations, medical licensing verifications, and compliance with data privacy acts.

10 Marketing and Go-to-Market Strategy

1. Key Strategies

- **Digital Outreach:** Engage regional medical practitioners and hospital managers on LinkedIn, medical boards, and clinical sub-reddits. We share technical case studies demonstrating how Sanjeevni reduces wait times and paperwork.
- **Strategic Partnerships:** Partner with local pharmacy cooperatives, rural NGOs, and medical colleges to build institutional trust. These collaborations facilitate direct onboarding pipelines.
- **Ambassador Programs:** Offer 14-day free trials and onboarding support through local health agents. This program builds trust among rural practitioners.
- **SEO Optimization:** Focus search ranking strategies on key terms like 'Safe Rural Telemedicine', 'Generic Medicine Lookup', and 'Visual Prescribing'. This captures organic traffic from clinic managers searching for solutions.

2. Customer Acquisition Plan

- **Phase 1:** Launch pilot integrations in 100 clinics through public health networks. This initial cohort verifies triage accuracy and translation engines in real-world scenarios.
- **Phase 2:** Onboard 500 users through regional cooperative pharmacy networks. This expansion builds a density of generic stock mappings across neighboring districts.
- **Phase 3:** Scale to 2,000+ clinical connections by integrating with state-sponsored public health initiatives and municipal health councils.

11 Funding and Support Needed

1. Funding Requirements

Sanjeevni seeks an initial development investment of ₹5–7 Lakhs for engineering, cloud databases, and clinic pilot testing. This seed funding will sustain operations through MVP launch, onboarding training, and initial revenue validation.

2. Use of Funds

- **MVP Software Development:** Consumes 40% of capital allocation. This covers core coding, React/Next.js interface building, PostgreSQL schema setups, and payment gateway hook integrations.
- **Field Trials & Training:** Accounts for 25% of capital allocation. This supports local travel, physical onboarding kits, regional training sessions, and user feedback collections.
- **Model & Cloud APIs:** Represents 20% of capital allocation. This funds cloud storage, server hosting, and API tokens for speech transcription and clinical record verification.
- **Compliance & Setup:** Consumes 15% of capital allocation. This funds security certifications, ABDM sandbox compliance audits, and business registration setups.

12 Social and Economic Impact

- **Health Democratization:** Enables remote patients in underserved districts to receive accurate pre-triage classifications and visual prescription cards. This ensures that non-literate patients can manage their dosages safely, reducing drug administration errors.
- **Socio-economic Relief:** Reduces travel expenses and consultation fees for rural families by bringing primary care clinics closer to home. This prevents health concerns from escalating into expensive emergency treatments.
- **Supply Chain Security:** Integrates retail pharmacies into a shared database network to locate generic compound substitutes when brand-name drugs are out of stock, ensuring treatment continuity.
- **Digital India Alignment:** Strengthens state health infrastructure in accordance with the Ayushman Bharat Digital Mission (ABDM), building structured medical record systems for emerging regions.

13 Risk Analysis

1. Potential Risks

- **Clinical Hallucinations:** Large Language Models are prone to fabricating medical advice, generating incorrect dosing recommendations, or misinterpreting symptoms. If raw AI outputs are delivered directly to patients or pharmacists, it could result in incorrect drug administration, creating severe medical safety issues and legal liabilities for the platform.
- **Low Clinic Adoption:** Rural pharmacists and clinic administrators are accustomed to manual paper books and simple ledger workflows. They may resist shifting to digital stock management due to unfamiliarity with technology or fear of complexity, which could delay platform onboarding and limit network density.
- **Data Security Breaches:** Telehealth files and clinical diagnostic records contain sensitive patient data. Without strict security controls, unauthorized record downloads or interception of files on insecure networks could leak private patient records, compromising patient privacy and violating regional data compliance policies.

2. Mitigation Strategies

- **Input Filters & Zod Validation:** We implement a multi-stage safety wrapper that scans symptoms against regex lists and filters out non-clinical topics. LLM outputs are parsed through Zod schema validators that enforce strict formatting rules. If a model output fails verification, the platform defaults to deterministic templates, protecting patients from hallucinations.
- **Onboarding Support:** We address adoption resistance by providing simplified, icon-based user interfaces and offline training sessions. Our team coordinates onboarding with local pharmacy managers to walk them through the system setup, building trust and ensuring smooth operations.
- **Credential-Verified Proxies:** All patient records and file downloads are routed through a credential-verified backend proxy gateway. The system uses Better Auth middleware to verify role-based permissions before decanting files, ensuring that only authorized medical personnel can access patient histories.

14 Scalability and Future Growth

- **Auto-scaling Infrastructure:** To handle sudden diagnostic requests and patient volume spikes, the platform is hosted on auto-scaling cloud infrastructure. The static client-side frontend is distributed via Vercel's CDN, and the backend Hono API engine runs on Render container clusters. This allows the system to scale resources dynamically during peak consultation hours, maintaining fast response times.
- **Modular Supply API integrations:** Sanjeevni includes modular inventory endpoints that allow drug distributors to update stock availability directly. This automates supply chain mapping, enabling village pharmacies to link their inventory with regional warehouses to reduce stock-out events and secure medication pipelines.
- **Global Expansion:** While pilots launch in northern Indian districts, the underlying prompt templates and translation pipelines are designed for global scalability. By Year 3, we plan to adapt translation modules and regional scripts

to expand the platform into emerging healthcare markets across Southeast Asia and East Africa.

15 Conclusion

Sanjeevni AI represents a paradigm shift in rural clinical care, offering a safe, highly resilient, and safety-aligned telehealth operating system specifically optimized for low-bandwidth and medicine-scarce environments. By replacing disjointed paper ledgers and fragmented communication channels with a single, unified digital workspace, Sanjeevni AI directly resolves systemic rural health bottlenecks including severe physician deficits, overwhelming medical triage backlogs, and administrative overhead burdens. Through innovative capabilities such as local client-side offline triaging, acoustic cough diagnostics, spatiotemporal epidemiological outbreak alert grids, and visual scheduling timeline cards, the platform guarantees that high-quality, multilingual clinical assistance remains accessible to rural families regardless of local connectivity drops or literacy levels.

Crucially, Sanjeevni AI mitigates the safety liabilities of off-the-shelf generative AI wrappers through a robust multi-stage safety pipeline, combining strict unicode validations and Zod output schema validations with automated generic drug compound lookup matching to eliminate hallucination risks. This proposal to the Concept2Action AI Innovation Incubator (AIII Program) seeks internship support to develop, pilot, and deploy the Sanjeevni AI MVP across targeted regional clinic pharmacy sub-networks. Supported by Concept2Action's structured mentorship, cloud database hosting resources, and public health sales networks, Sanjeevni AI will scale to establish a safe, reliable, and compliant digital medical operating standard for emerging communities worldwide.

Sanjeevni AI — The Safe, Resilient Operating System for Rural Telehealth.

16 References

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